

ROGRAKSHAK SYNTHETIC MEDICAL CENTER

DEPARTMENT OF PATHOLOGY & CLINICAL MICROBIOLOGY
Diagnostic Services Wing, Central Tower | Tel: +91 (22) 555-0199 | NABL & CAP Accredited Laboratory

LABORATORY MICROBIOLOGY & ANTIMICROBIAL SUSCEPTIBILITY REPORT

Patient Name:	Nisha Chopra	Specimen ID:	ACC-2026-00013
MRN:	MRN-2026-1014	Order ID:	ORD-MIC-1013
Patient ID:	PAT-0014	Specimen Type:	Sputum
Age / Sex:	43 Y / Female	Collected Date/Time:	2026-08-07 18:00
Admitting Diagnosis:	COPD Exacerbation	Received Date/Time:	2026-08-07 18:45
Report Status:	FINAL	Reported Date/Time:	2026-08-09 22:00

I. PRIMARY SPECIMEN EXAMINATION & CULTURE FINDINGS

Direct Smear / Microscopy:	Gram-negative bacilli observed (moderate to heavy). Polymorphonuclear leukocytes seen.
Culture Growth Result:	Significant Pure Growth: > 10 ⁵ CFU/mL isolated on selective and non-selective media.
Isolated Organism:	<i>Pseudomonas aeruginosa</i>
Resistance Classification:	SUSCEPTIBLE
Identification Methodology:	Automated Microbial Identification (VITEK 2 GN/GP & MALDI-TOF Mass Spectrometry)

II. ANTIMICROBIAL SUSCEPTIBILITY TESTING (AST) — CLSI M100 CRITERIA

Antimicrobial Agent	MIC / Value	Interpretation	CLSI Category
Piperacillin / Tazobactam	<= 8/4 ug/mL	Susceptible (S)	S
Ceftazidime	<= 2 ug/mL	Susceptible (S)	S
Cefepime	<= 2 ug/mL	Susceptible (S)	S
Meropenem	<= 1 ug/mL	Susceptible (S)	S
Imipenem	<= 1 ug/mL	Susceptible (S)	S
Ciprofloxacin	<= 0.25 ug/mL	Susceptible (S)	S
Levofloxacin	<= 0.5 ug/mL	Susceptible (S)	S
Amikacin	<= 4 ug/mL	Susceptible (S)	S
Tobramycin	<= 1 ug/mL	Susceptible (S)	S

III. INTERPRETATION & CLINICAL REMARKS

Interpretation: Isolated strain of *Pseudomonas aeruginosa* exhibits standard wild-type susceptible antimicrobial profile. No high-level or extended-spectrum acquired resistance detected. Therapy should be directed based on documented clinical susceptibility and site of infection.

IV. LABORATORY AUTHORIZATION & AUTHENTICATION

Laboratory Report ID: LAB-0013	Testing Lab: Division of Clinical Microbiology	Status: FINAL
Medical Technologist: S. Nambiar, MSc (Med Micro) Reg #MLT-2021-884	Reviewing Pathologist: Dr. Arvind Deshmukh, MD (Path) Reg #MMC-2012-0498	Consultant Microbiologist: Dr. Ramesh Kulkarni, MD, DNB Verified: 2026-08-09 22:00